Z, represent the number of protons and the ordinates, $N-Z$, the amount by which the number of neutrons exceeds that of the protons. This is the method used in our Tables IVa and IVb. The order of the individual elements also corresponds to their sequence in the periodic system. In the tables, the nuclei are also distinguished according to their stability or radiation characteristics. Stable nuclei are represented by black dots. One of these is the nucleus ${}_2\mathrm{He}^4$, which appears where $Z = 2$ and $N-Z = 0$. Triangles indicate the radioactive nuclei which emit beta rays. When the apex of the triangle points upward, it indicates an emitter of electrons; a triangle with the apex pointing downward indicates an emitter of positrons. The electron emitters always appear in the uppermost row—e.g., ${}_2\mathrm{He}^6$ or ${}_3\mathrm{Li}^8$—and the positron emitters mostly in the lowest row, e.g., ${}_6\mathrm{C}^{11}$. The radioactive nuclei which emit alpha particles, are indicated by small squares. Finally, there are unstable nuclei which change by the capture of an electron from the innermost extranuclear electron shell and thus reduce their nuclear charge number by one unit. In our tables, these nuclei are indicated by small circles. Those nuclei which can emit both electrons and protons, are marked by two triangles superimposed on each other in the shape of a star. Thus, our tables furnish a simple survey of all the existing nuclei, their structures and properties.

As we see, the neutron excess, $N-Z$, is negative in just a few species of atoms only; in all the rest, it is a positive magnitude, and never a very big one. The nuclei of the lighter elements contain, throughout, practically as many neutrons as they contain protons, and the neutron excess becomes more or less considerable in the heavier elements only.

We have now described, in general outline, the structure of individual nuclei, and this discussion brings up a great many further questions: What is holding together a nucleus which consists of protons and neutrons? What is the nature of the forces that bind these particles together? Why do lighter nuclei contain approximately as many protons as neutrons, while the heavier ones show a slight neutron excess—in other words, why does the magnitude of the neutron excess increase with the increase of the number of unit charges on the nucleus? Why is there only a limited number of nuclei? Why are many of these radioactive, and why do they emit those very particles which we observe to be emitted by them? In the following lectures, we shall discuss these problems.